GridPulse

Energy Community Intelligence

Optimize the Flow

Executive Summary

Investor & Partner Briefing

Pure SaaS orchestration for Energy Communities

No hardware. · No retrofits. · Real savings.

2026 · Strictly Confidential

Founding Team

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Energy that works smarter — together.

GridPulse is a pure-software SaaS platform that transforms fragmented household energy assets into a coordinated, AI-optimised Energy Community. No hardware. No retrofits. Fully compliant with EU RED III. Built for scale across European markets.

THE PROBLEM

European electricity markets are structurally volatile. Day-ahead spot prices (EPEX) swing from **–€50/MWh** during midday solar overproduction to **+€200/MWh** during evening demand peaks — the so-called **duck curve**. Residential consumers, even those with PV, EVs, or batteries, remain price-blind: they neither forecast these swings nor act on them in real time.

The Energy Community paradox. EU Directive 2018/2001 (RED II) and its 2023 revision (RED III) grant citizens the legal right to form Energy Communities — collective structures that can trade electricity internally, aggregate flexibility, and receive preferential grid tariffs. Yet adoption is near-zero. Our D3 validation ($n=9$) confirmed: **0 of 9 respondents were EC members**. The barrier is not technology literacy — all nine grasped the **GridPulse** concept after a short demo. The barrier is **regulatory complexity and absent orchestration tooling**.

The asset silo problem. Households increasingly own PV, EV chargers, heat pumps and batteries — controlled by separate apps with no awareness of neighbours or live market prices. The result: missed arbitrage, near-zero feed-in tariffs, and avoidable peak bills.

WINDOW OF OPPORTUNITY

Regulatory tailwind — RED III. The revised Renewable Energy Directive mandates that all EU member states establish enabling frameworks for Energy Communities by end of 2025. National transposition is creating a **first-mover window**: EC operators, housing associations and municipalities are actively seeking management solutions to comply, and the competitive field for pure-software orchestration remains uncrowded.

Asset density crossing the threshold. The EC use case becomes compelling at sufficient density of flexible assets. European residential electrification is crossing that threshold now: EU EV sales exceeded 1.5M units in 2023; heat pump installations grow at 12% CAGR; residential BESS is the fastest-growing distributed energy segment. **GridPulse** enters precisely as the asset base reaches viable EC density.

DSO grid stress — a second revenue vector. DSOs face increasing LV grid congestion as unmanaged EV and heat pump peaks stress transformer capacity. **GridPulse**'s community-level load orchestration and its future **DSO LV grid health monitoring module** positions the platform as a grid service provider, opening B2B DSO contracts alongside the B2B2C community business.

Lock-in dynamics. Once deployed, **GridPulse** accumulates asset integration data, optimisation history and member trust — structurally high switching costs creating **durable retention and predictable ARR**.

THE SOLUTION

GridPulse is a hardware-agnostic SaaS intelligence layer connecting via open APIs to PV inverters, EV charge points, heat pumps, BESS and boilers, orchestrating them as a single optimised Energy Community.

Four-step real-time control loop:

- **Market Signal Ingestion:** EPEX day-ahead and intraday prices ingested continuously with live grid carbon intensity tracking.
- **2-Hour Rolling ML Forecast:** Integrates weather, PV yield, load history, battery SoC and price trajectory to pre-position assets before price events occur.
- **Sub-100ms API Commands:** Optimised dispatch sent simultaneously to all controllable assets — charge, hold, discharge, shift, export — with full auditability and a continuous feedback loop.
- **Outcome & Feedback Loop:** Per-household and EC-level savings logged. Actuals feed back into the model, continuously improving forecast accuracy and dispatch quality.

Asset-manufacturer-agnostic. No new hardware. Remote deployment. A 50-household EC can be onboarded within days of contract signature.

MARKET OPPORTUNITY

€18.4B
Total Addressable Market

Full European multi-family residential & commercial building stock requiring advanced energy management under EU decarbonisation mandates.

€3.2B
Serviceable Addressable Market

Hardware-agnostic software orchestration for EU RED III compliance. Initial focus: DACH, Spain & France — highest EC regulatory maturity.

€145M
Serviceable Obtainable Market (5yr)

4.5% penetration across municipal housing networks, real-estate co-proprietor boards & industrial clusters within five years.

WHY NOW

- RED III compliance deadlines active EU-wide
- No dominant pure-software EC orchestration player
- EV & HP penetration at EC-viable asset density
- DSOs actively procuring demand flexibility solutions
- Day-ahead spot volatility at multi-year highs

PRODUCT & COMPETITIVE DIFFERENTIATION

KEY DIFFERENTIATOR: COMMUNITY-LEVEL MULTI-HOUSEHOLD ORCHESTRATION

The fundamental architectural distinction of **GridPulse** is its **community-first dispatch logic**. Single-home HEMS solutions optimise one household in isolation, blind to neighbouring assets. **GridPulse** maintains a **real-time community-wide energy model**: surplus PV from one household charges the EV of another; a pure consumer with no solar assets still benefits by purchasing community electricity at internal trading prices materially below the retail grid tariff; EVs, hot-water tanks and battery systems function as **community-level virtual buffers**, absorbing cheap midday generation and releasing it during expensive peaks for all members. This transforms an Energy Community from a regulatory entity into an economically self-reinforcing system — and creates a defensible **network effect** that single-home solutions structurally cannot replicate.

COMPETITIVE LANDSCAPE — POSITIONING MATRIX

Solution	Community-wide orchestration	Hardware-free	Price-adaptive	EC-native	DSO grid services
GridPulse	**✓ Full EC dispatch**	**✓ Pure SaaS**	**✓ Sub-100ms**	✓	**✓ LV health monitor**
Single-home HEMS (Loxone, Viessmann)	× Single unit only	✓	~ Partial	×	×
Energy Retailers (Tibber, Octopus)	~ Tariff signals only	✓	~ Fixed dynamic tariff	×	×
Hardware Bundlers (SMA, SolarEdge)	~ Own-device ecosystem	× Proprietary lock-in	~ Partial	×	×
VPP Aggregators (Next Kraftwerke)	~ Portfolio-level only	✓	✓	×	~ Wholesale only

BUSINESS MODEL & VALIDATED ASSUMPTIONS

REVENUE MODEL & UNIT ECONOMICS

GridPulse operates a **25% revenue-share model** on verified, metered household savings — fully aligning platform incentives with operator and member outcomes. **GridPulse** earns only when savings are delivered and independently verified.

Go-to-market: B2B2C whole-EC licensing. GridPulse contracts with EC operators — housing associations, municipal utilities, real-estate managers — who deploy to all households including pure consumers. D3 validated: **8/9 preferred whole-EC licensing** over per-household subscription, citing administrative simplicity.

Unit economics (50-household EC):

- Avg. gross saving per household: **€400/year**
- **GridPulse** revenue share (25%): **€100/HH/year**
- Community ARR: **€5,000/community/year**
- At 1,000 ECs (50k HH): **€5M ARR**

A second revenue stream activates from 2028 via **DSO grid service contracts** (FCR, FRR, congestion management) — high-margin and non-correlated with consumer SaaS revenue.

SAVINGS MECHANICS — 6 STACKED LEVERS

GridPulse compounds savings across six independent additive mechanisms:

- **Dynamic tariff optimisation:** Shift flexible loads into negative or low-price windows. Est. 12% of annual consumption cost at baseline volatility.
- **PV self-consumption:** Route surplus within community before grid export. Raises self-consumption from ∼45% baseline to ∼67%+.
- **Internal P2P trading:** Sellers earn above feed-in tariff; buyers pay below retail rate. Spread stays within the community.
- **Peak demand reduction:** Suppress coincident peaks triggering capacity-based grid tariff components. Est. €45/HH/year.
- **EV smart charging:** Charge at highest renewable share and lowest spot price windows. Est. €110/EV/year.
- **BESS arbitrage:** Buy-low/discharge-high cycle optimisation across the community battery fleet. Est. €65/BESS/year.

Aggregate outcome: up to 30% reduction in household energy costs.

Metered pilot evidence targeted Q1 2027.

D3 PRIMARY MARKET VALIDATION

Nine structured in-depth interviews with prospective EC members, property managers, and energy-aware consumers (DACH region) under a Business Model Canvas (BMC) validation methodology.

- **9/9** grasped the **GridPulse** concept after a live demo — confirming the **demo-first acquisition strategy**.
- **0/9** were current EC members. Dominant barrier: **regulatory opacity and absent tooling** — not tech aversion.
- **8/9** prefer **whole-EC B2B2C licensing** over per-household subscription.
- **Revenue-share** outperforms flat-fee in all willingness-to-pay assessments.
- **5/9** require pilot data to fully accept the 30% savings claim — the primary conversion accelerator.

Strategic implication: PMF confirmed at concept level. Acquisition bottleneck is regulatory navigation and pilot credibility — both addressable in the Q4 2026–Q1 2027 validation phase.

STRATEGIC ROADMAP

Q4 2026	Q1 2027	Q3 2027	Q4 2027	2028	2030
Final Validation	**Algorithm Testing**	**Deploy GridPulse**	**5,000 Customers**	**EU Regional Scale**	**VPP Platform**
Evaluate most efficien EC approach across Eu ropean countries. F nalise optimal structures in DACH, Spain & Franc Confirm B2B2C operato pipeline.	First live pilot run on a real EC. Optimise contro algorithm. Generate me tered savings data to sub stantiate the 30% claim.	Targeted deployment in 5–10 large-scale ECs Repeatable sales & on boarding workflow. Firs metered ARR generated	5,000 household sale across active ECs Proven cost reduction a residential scale. GTN motion standardised.	Secondary rollouts France, Spain, Benelu & UK. DSO service con tracts initiated. Series A fundraising readiness.	Full VPP capability. DSO LV grid health state checks. EU capacity market participation: FCR & FRR.

FINANCIAL OUTLOOK (ILLUSTRATIVE)

Revenue model: 25% share of verified household savings.

Phase	Target ARR
Q4 2027	€250k (5–10 ECs, ~500 HH)
2028	€1.5M (EU regional rollout)
2030	€5M+ (VPP + DSO revenue)

Key SaaS metrics targets:

- Net Revenue Retention >110% (asset density grows within live ECs)
- Gross Margin >75% (pure software, near-zero marginal cost per HH)
- CAC payback <12 months via B2B2C operator channel

FOUNDING TEAM

Jagruthi Nadendla · Co-founder
Stuti Srivastava · Co-founder
Vaishnavi Javvaji · Co-founder
Jakob Erd · Co-founder

Four-person founding team combining expertise in energy systems engineering, machine learning, regulatory policy, and business development. Built and validated at the intersection of European energy market reform and AI-driven optimisation. Active engagement with EC operator networks, DSO representatives, and energy regulatory bodies across DACH and wider EU.

WHAT WE ARE LOOKING FOR

Pilot Partners
EC operators, housing associations, or municipalities prepared to run a live **GridPulse** deployment in Q1 2027. Pilots are fully managed remotely — no internal technical resource required from the operator.

Seed & Pre-Seed Investors
Early-stage investors aligned with the European energy transition, regulatory SaaS, and infrastructure software. Targeting funding through Q4 2026 final validation and Q1 2027 pilot phase to first metered ARR.

Strategic Partners
Inverter OEMs, EV charge point operators, heat pump manufacturers and DSOs seeking a demand-flexibility and grid-service partner. API integration partnerships accelerate asset coverage and route-to-market.

Contact: **gridpulse.io**

Disclaimer: Forward-looking statements and market projections prepared in good faith. Savings figures are modelled projections; actual results vary by location, tariff, asset mix and utilisation. Metered pilot evidence for the 30% savings claim targeted Q1 2027. TAM/SAM/SOM derived from EU household and energy market data. Financial projections are illustrative only. Confidential — not for redistribution without written consent. © 2026 GridPulse.